PAPER_1558_M_TAU_1_777

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PAPER_1558 — Tau Lepton Mass = $m_\tau = SSQ + F_{TRZ} \cdot D + F_{TRZ} \cdot SO_5 - F_{TRZ}^2 \cdot D_{crit} + F_{TRZ}^2 \cdot D_{BSFG} + F_{TRZ}^2 \cdot (SSQ + SSQ^2 - SSQ^3) \approx 1.777$ GeV

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Particle Physics
Date: June 18, 2026
Status: CLOSED — 0.013% closure from PAPER_1209HH_S659

Observation

PAPER_1209HH_S659 (Particle Physics Unified Proof Set) lists **Tau Lepton Mass** with the integer-primitive expression:

```
Tau Lepton Mass = m_τ = SSQ + F_TRZ·D + F_TRZ·SO_5 − F_TRZ²·D_crit +
F_TRZ²·D_BSFG + F_TRZ²·(SSQ + SSQ² − SSQ³) ≈ 1.777 GeV
```

Status: **0.013%**.

UQFF Closed Identity

```
m_τ = SSQ + F_TRZ·D + F_TRZ·SO_5 − F_TRZ²·D_crit + F_TRZ²·D_BSFG + F_TRZ²·(SSQ
+ SSQ² − SSQ³) ≈ 1.777 GeV 0.013%
```

Physical Interpretation

Heaviest charged lepton. Integer-primitive expansion starting from SSQ = 0.57 and adding F_TRZ-modulated integer-primitive corrections. Matches CODATA 1.77686 GeV at 0.013%.

NOT REPLACEMENT

CODATA/PDG treats Tau Lepton Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S659
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_tau_1_777"})`
